MODEL LAB · 2026-10-27

Every prior, same slate

The held-out scores that decided which prior is served, then how each one would have ranked today's hitters.

Served (full season)
served prior

11
priors compared

17,710
held-out plate appearances

2.92%
held-out HR rate

0.6014
best AUC · KNN comps

Held-out scores

Fitted on plate appearances before 2026-09-13, scored on the 17,710 that came after (actual HR rate 2.92%). No outcome in the test window was visible during fitting. **Read AUC first** — the slate ranks hitters and takes the top six, so ordering is what matters. Brier and log loss are dominated by the ~3% base rate and barely separate the priors. Metrics are per plate appearance; the probabilities lower down are per game.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.5820	1.61×	0.02829	0.13108	0.00466
KNN comps Prior is the pooled rate of the nearest comparables.	0.6014	1.64×	0.02824	0.12993	0.00614
SVR Support-vector regression on the same feature matrix.	0.5940	1.51×	0.02829	0.13054	0.00689
Random forest Forest regression; a learned similarity metric.	0.5967	1.51×	0.02828	0.13022	0.00676
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.6014	1.64×	0.02824	0.12993	0.00614
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.6014	1.64×	0.02824	0.12993	0.00614
Linear regression Ordinary least squares on the same features.	0.5979	1.57×	0.02828	0.13031	0.00488
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5969	1.59×	0.02833	0.13060	0.00615
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5971	1.55×	0.02827	0.13021	0.00494
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5961	1.45×	0.02832	0.13060	0.00718

BLEND WEIGHTS (FOREST VARIANT)

KNN **1.00** SVR **0.00** RF **0.00**

SHRINKAGE K PER PRIOR

league **260** knn **260** svr **260** rf **260** core **260** ensemble **260** linear **260** logistic **260** nn **260** xgb **260**

Estimated by method of moments on out-of-fold residuals, not hand-tuned.

Thin-sample subset

The same held-out window, restricted to hitters with fewer than 100 plate appearances before the cutoff (1,620 PA, actual HR rate 2.35%). For a regular with 600 PA the observed record swamps any prior, so the priors mostly separate here.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.6456	1.58×	0.02291	0.11126	0.01790
KNN comps Prior is the pooled rate of the nearest comparables.	0.6062	1.58×	0.02288	0.10980	0.01007
SVR Support-vector regression on the same feature matrix.	0.5783	1.58×	0.02323	0.11613	0.02309
Random forest Forest regression; a learned similarity metric.	0.6000	1.32×	0.02320	0.11319	0.02260
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.6062	1.58×	0.02288	0.10980	0.01007
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.6062	1.58×	0.02288	0.10980	0.01007
Linear regression Ordinary least squares on the same features.	0.6112	1.32×	0.02322	0.11383	0.02025
Logistic regression Binomial logistic on the per-PA home run outcome.	0.6015	1.32×	0.02354	0.11442	0.02384
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.6056	1.58×	0.02300	0.11128	0.01557
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5934	1.05×	0.02361	0.11640	0.02762

Slate agreement

PRIOR	SHARED	ITS TOP SIX
Served (full season)	6/6	Jackson Chourio 17.0%, Junior Caminero 15.5%, William Contreras 15.3%, Jake Bauers 15.1%, Christian Yelich 13.1%, Jonathan Aranda 12.2%
Flat league	5/6	Jackson Chourio 17.3%, William Contreras 15.1%, Junior Caminero 14.8%, Jake Bauers 14.0%, Christian Yelich 13.1%, Yandy Díaz 12.4%
KNN comps	6/6	Jackson Chourio 17.6%, Junior Caminero 15.3%, William Contreras 15.2%, Jake Bauers 15.2%, Christian Yelich 13.4%, Jonathan Aranda 12.3%
SVR	5/6	Jackson Chourio 17.9%, Junior Caminero 16.6%, William Contreras 15.2%, Jake Bauers 15.0%, Christian Yelich 12.7%, Yandy Díaz 12.6%
Random forest	5/6	Jackson Chourio 17.8%, Junior Caminero 16.0%, Jake Bauers 15.3%, William Contreras 15.2%, Christian Yelich 13.1%, Yandy Díaz 12.4%
Core (KNN+SVR)	6/6	Jackson Chourio 17.6%, Junior Caminero 15.3%, William Contreras 15.2%, Jake Bauers 15.2%, Christian Yelich 13.4%, Jonathan Aranda 12.3%
Forest blend	6/6	Jackson Chourio 17.6%, Junior Caminero 15.3%, William Contreras 15.2%, Jake Bauers 15.2%, Christian Yelich 13.4%, Jonathan Aranda 12.3%
Linear regression	6/6	Jackson Chourio 17.8%, Junior Caminero 15.5%, William Contreras 15.2%, Jake Bauers 15.0%, Christian Yelich 13.3%, Jonathan Aranda 12.3%
Logistic regression	5/6	Jackson Chourio 17.8%, Junior Caminero 15.9%, William Contreras 15.2%, Jake Bauers 15.0%, Christian Yelich 12.8%, Yandy Díaz 12.1%
Neural net (torch)	5/6	Jackson Chourio 17.8%, Junior Caminero 15.9%, William Contreras 15.3%, Jake Bauers 15.2%, Christian Yelich 12.9%, Yandy Díaz 12.2%
XGBoost	5/6	Jackson Chourio 18.0%, Junior Caminero 16.2%, Jake Bauers 15.3%, William Contreras 15.2%, Christian Yelich 12.9%, Yandy Díaz 12.5%

Hitter by hitter

The top 40 bats by the served probability.

HITTER	SERVED (FULL SEASON)	FLAT LEAGUE	KNN COMPS	SVR	RANDOM FOREST	CORE (KNN+SVR)	FOREST BLEND	LINEAR REGRESSION	LOGISTIC REGRESSION	NEURAL NET (TO
1 Jackson Chourio MIL slate	**17.0%**	17.3%	17.6%	17.9%	17.8%	17.6%	17.6%	17.8%	17.8%	
2 Junior Caminero TB slate	**15.5%**	14.8%	15.3%	16.6%	16.0%	15.3%	15.3%	15.5%	15.9%	
3 William Contreras MIL	**15.3%**	15.1%	15.2%	15.2%	15.2%	15.2%	15.2%	15.2%	15.2%	
4 Jake Bauers MIL	**15.1%**	14.0%	15.2%	15.0%	15.3%	15.2%	15.2%	15.0%	15.0%	
5 Christian Yelich MIL	**13.1%**	13.1%	13.4%	12.7%	13.1%	13.4%	13.4%	13.3%	12.8%	
6 Jonathan Aranda TB	**12.2%**	11.6%	12.3%	11.9%	12.0%	12.3%	12.3%	12.3%	12.0%	
7 Yandy Díaz TB	**12.0%**	12.4%	12.1%	12.6%	12.4%	12.1%	12.1%	12.2%	12.1%	
8 Brice Turang MIL	**11.7%**	11.9%	11.4%	11.6%	11.3%	11.4%	11.4%	11.0%	10.9%	
9 Garrett Mitchell MIL	**10.7%**	10.2%	10.6%	9.9%	10.3%	10.6%	10.6%	10.1%	10.1%	
10 Ryan Vilade TB	**10.6%**	9.9%	10.4%	10.1%	10.5%	10.4%	10.4%	10.1%	9.9%	
11 Jonny DeLuca TB	**8.2%**	8.8%	7.8%	7.7%	7.4%	7.8%	7.8%	7.8%	8.1%	
12 Sal Frelick MIL	**7.5%**	9.3%	7.7%	7.2%	6.9%	7.7%	7.7%	7.5%	7.6%	
13 Liam Hicks TB	**7.3%**	8.2%	7.2%	8.0%	7.8%	7.2%	7.2%	7.6%	7.5%	
14 Cooper Pratt MIL	**7.0%**	9.0%	6.9%	6.6%	6.5%	6.9%	6.9%	5.9%	6.3%	
15 David Hamilton MIL	**6.6%**	7.9%	6.4%	6.0%	5.8%	6.4%	6.4%	6.3%	6.3%	
16 Richie Palacios TB	**5.6%**	6.3%	5.6%	5.3%	4.9%	5.6%	5.6%	5.3%	5.3%	
17 Taylor Walls TB	**3.9%**	5.1%	3.7%	2.9%	2.9%	3.7%	3.7%	3.6%	3.7%	
18 Chandler Simpson TB	**2.1%**	4.7%	2.8%	1.2%	1.4%	2.8%	2.8%	1.1%	2.4%	

Per-game probabilities: each prior's per-PA rate run through this hitter's own plate appearances, park, weather, bullpen and times-through-the-order terms. Only the prior changes across the columns — every other term is held fixed, so a left-to-right spread is the prior disagreeing about the hitter, not the context. Shading grows with the probability.

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